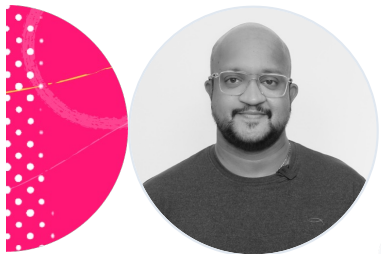


Deploying AI Agents in Production Environments

Preparing and Launching Your AI Agent



Pratheerth Padman

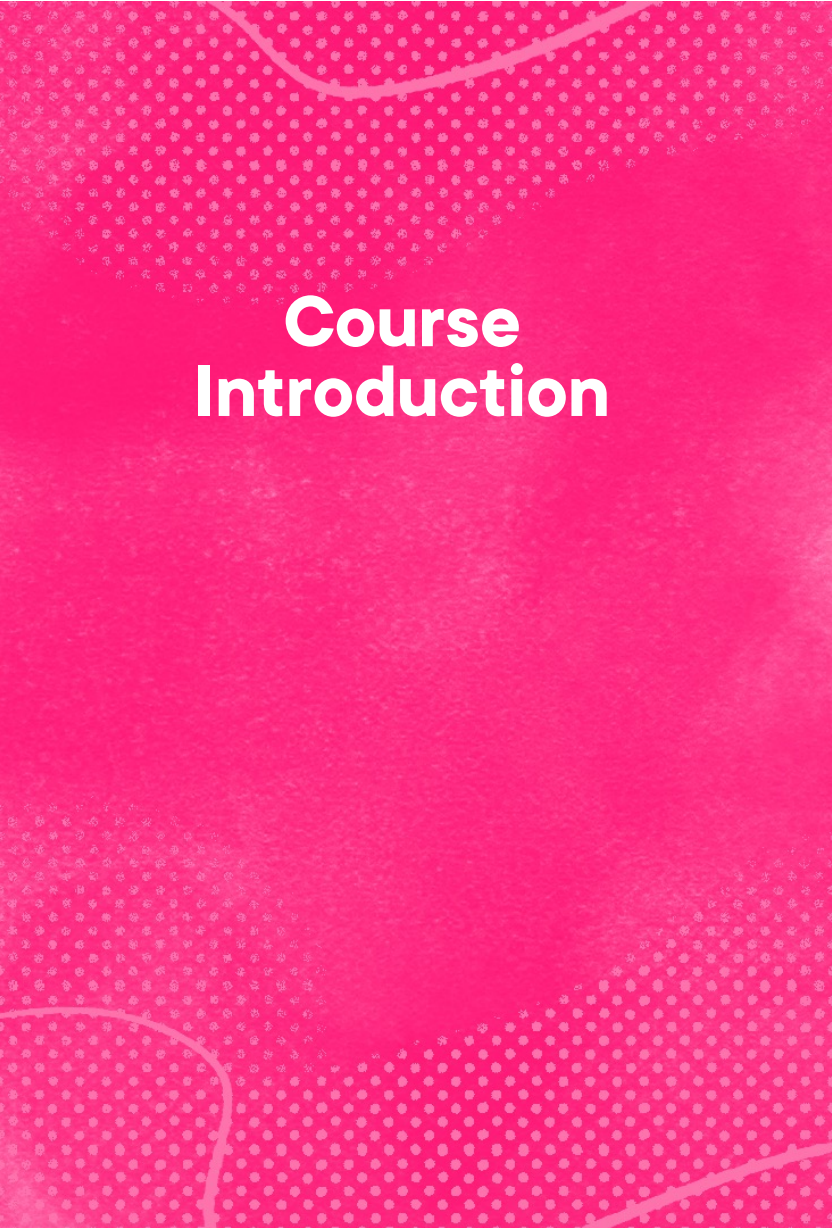
Machine Learning Engineer

@theaibug | www.pratheerthpadman.com



**How do you get your
agent out of your local
setup and into the real
world?**





Course Introduction

Preparing your agent for production

Launching your containerized agent

Monitor and maintain your deployed agent

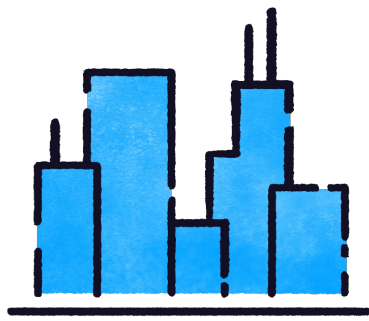




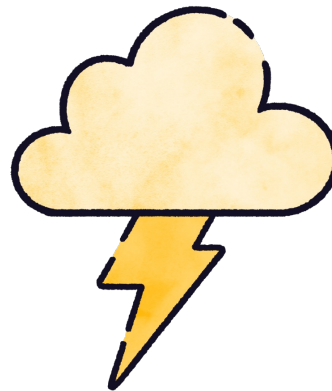
Our Agent's Journey to Production



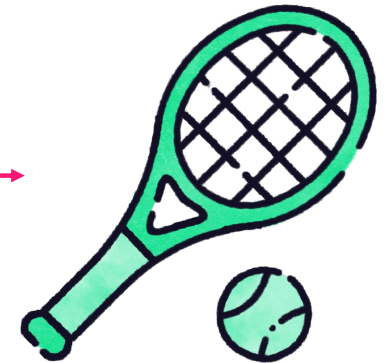
Activities Advisor Agent



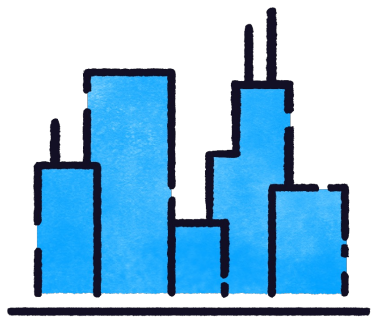
OpenWeather API



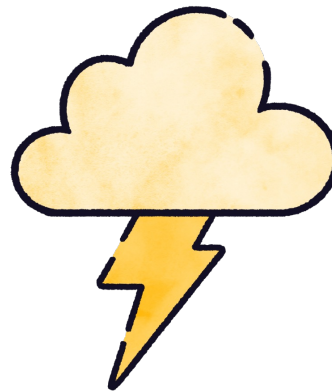
Tavily Search



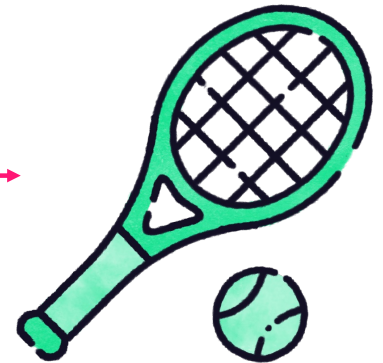
Activities Advisor Agent



OpenWeather API



Tavily Search



FastAPI

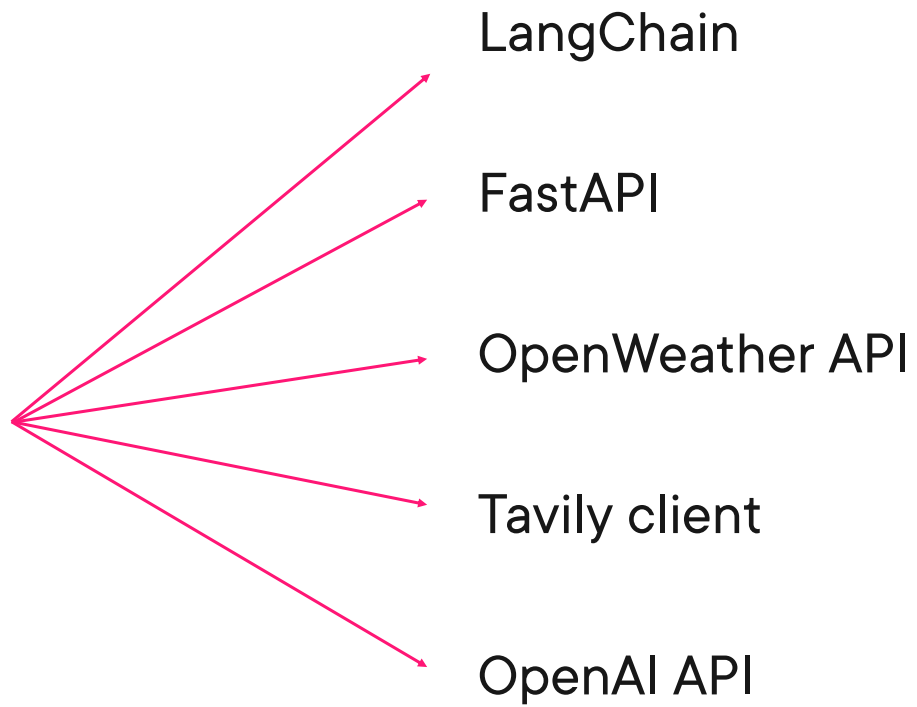
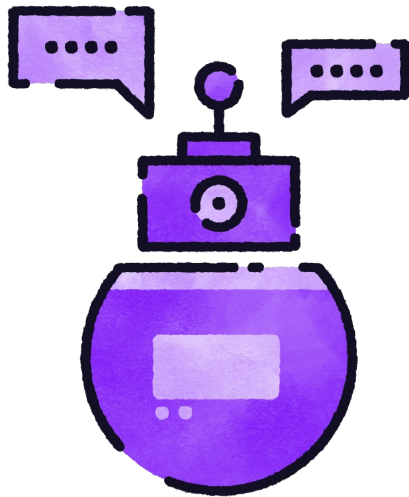




Anticipating Production Hurdles



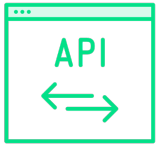
Dependencies



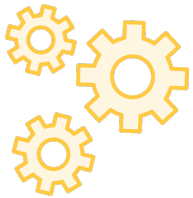
We need to ensure that the exact versions of these dependencies that work on our machine are replicated in the production environment.



Configuration Management



API keys



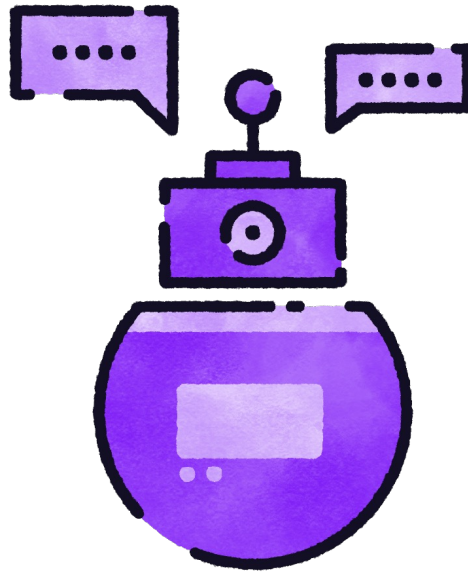
Settings for your agent in development versus production mode



Sensitive data needs protection – no hardcoding!



Scaling



Scaling



Security



Validating inputs



Securing credentials



Controlled access



Production Hurdles

Dependencies

Configurations

Scaling

Security





Choosing Your Cloud Deployment Strategy



Options for Cloud Deployment



Fully managed agent platforms



Full control environments



Serverless functions



Serverless container platforms



Specialized and advanced solutions



Two Components of Our Deployment Strategy

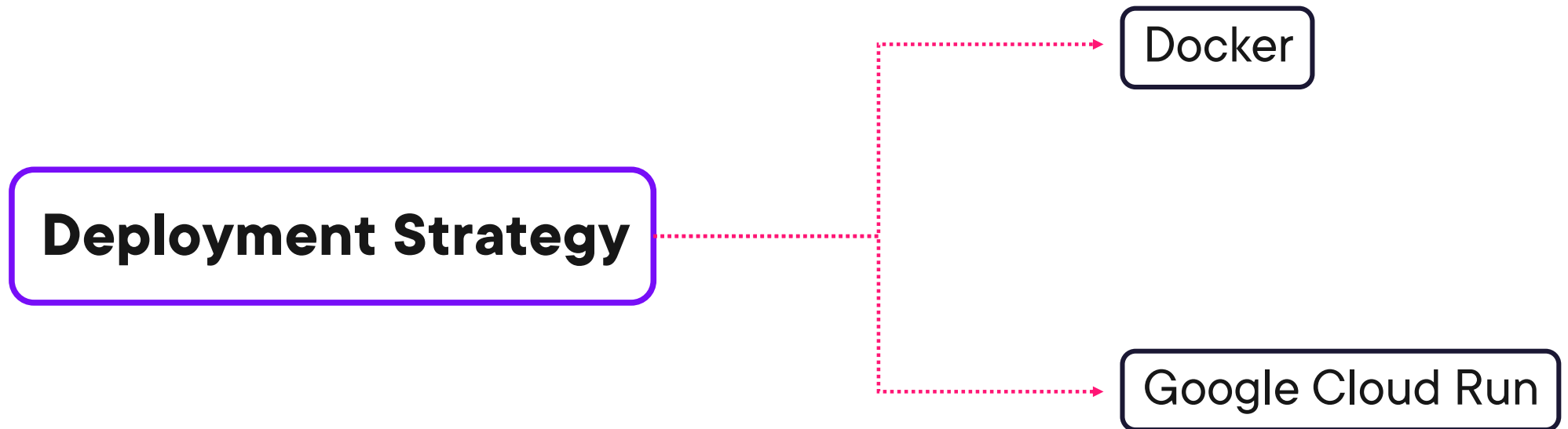


What Is Docker?

Docker allows us to package our application into a standardized unit called Container. This container has everything the agent needs to run, whether it's on our laptops or in the cloud.



Two Components of Our Deployment Strategy



What Is Google Cloud Run?

Cloud Run is a managed platform from Google Cloud that's designed to run stateless containers. It's 'serverless,' which means Google handles a lot of the underlying infrastructure management for us.



Why Docker + Cloud Run?



Docker is fantastic for ensuring our AI agent's environment is reproducible

Cloud Run lets us deploy containers easily, provides auto-scaling, and we don't have to manage servers

Cloud Run has a generous free tier and a pay-per-use model

You can configure CPU, memory, and min/max instances to balance performance and budget





Demo: Containerizing Your Agent with Docker

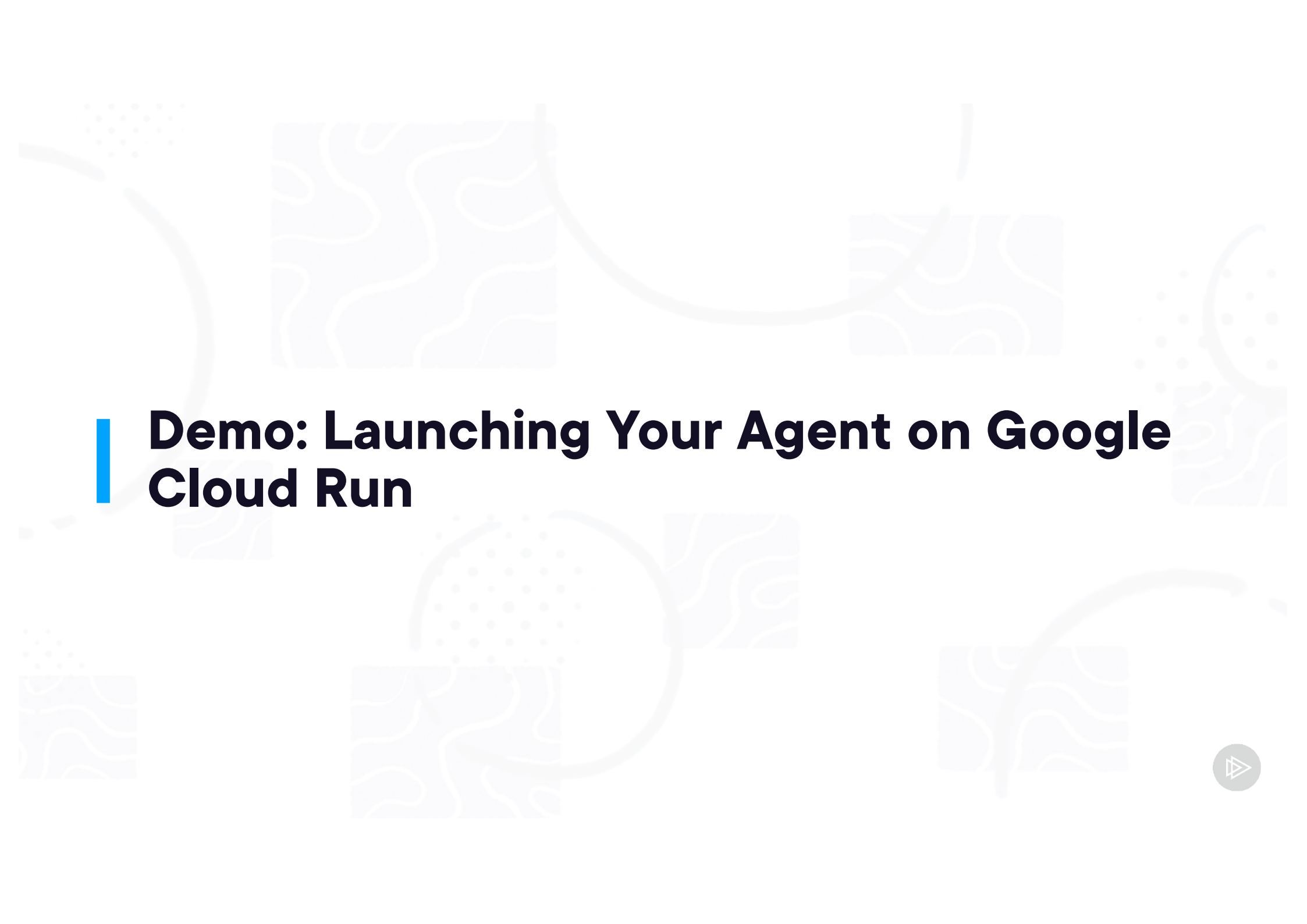




What Is Dockerfile?

Dockerfile is a simple text file that contains a set of instructions used to build a Docker image. Think of it like a recipe or blueprint that tells Docker how to create a customized container.





Demo: Launching Your Agent on Google Cloud Run

